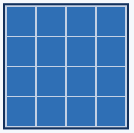
VECTOR SPACES · SUBSPACE SUMS

Sum and direct sum

One construction with one condition attached. Whether the intersection is trivial decides the dimension formula, the uniqueness of decompositions, and whether the notation $\oplus$ is permitted at all.

2
CASES

THE GENERAL CASE 1

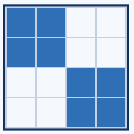


Sum | $W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2\}$

DIMENSION	INTERSECTION	DECOMPOSITION
$\dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$	may be nontrivial	exists, but not unique

The correction term is a double count: anything in both subspaces is added twice and must be removed once. Two planes through the origin in $\mathbb{R}^3$ have dimensions summing to four in a three-dimensional space — the intersection line is the missing one.

WHEN THE INTERSECTION IS TRIVIAL 1



Direct sum | $W_1 \cap W_2 = \{\mathbf{0}\}$

DIMENSION	INTERSECTION	DECOMPOSITION
$\dim W1 + \dim W2$	trivial — the origin only	exists and is unique

With nothing shared there is nothing to subtract, and each vector splits in exactly one way. This is the structure behind orthogonal complements and behind every basis: V is the direct sum of the lines spanned by its basis vectors.

The direct sum is not a different operation — it is the general sum with $W_1 \cap W_2 = \{\mathbf{0}\}$, at which point the correction term vanishes and every decomposition becomes unique. The two rows differ in one hypothesis and everything else follows from it.